Course - 1 (Quantum Mechanics and Quantum Information) (August 2026 - October 2026)

May, 2026- July, 2026: LA (ask Tanmoy to be a TA)

July-October: Internship program

September-November: QM and QI

November-January: DG

Works that need to be done:

1. Zoom recording and uploading to Youtube
2. Zoom turning on and being present for the whole duration
3. Homeworks: (video explaining a topic)

1. Walid makes poster for LA and he mentioned he is interested in making a website.

Preparatory program on Quantum Mechanics:

BASIC TOPICS:

(Arpit Das, Naveen Umasankar, (Subhroneel?))

Lecture 01: Motivation behind Quantum Mechanics - Statistical interpretation and Probability
Lecture 02: Normalization, Momentum, Uncertainty Principle
Lecture 03: Time-independent Schrodinger Equation, Stationary states and infinite square well
Lecture 04: The Harmonic Oscillator
Lecture 05: The free particle and delta function potential
Lecture 06: Introduction to Dirac Formalism - Hilbert space and the theory of observables.
Lecture 07: Eigenvalue and Eigenfunctions of Hermitian operators
Lecture 08: Generalized statistical interpretation and uncertainty principle

Lecture 09: Quantum mechanics in three-dimensions - Schrodinger equation in spherical coordinates
Lecture 10: Continued from Lecture 09
Lecture 11: The Hydrogen Atom - 1
Lecture 12: The Hydrogen Atom - 2
Lecture 13: Angular momentum
Lecture 14: Spin - 1
Lecture 15: Spin - 2
Lecture 16: Review of Time-independent Perturbation Theory
Lecture 17: Review of Time-dependent Perturbation Theory
Lecture 18: WKB Approximation

QUANTUM INFORMATION: (Sowmitra Das)

Lecture 19: Introduction to Quantum Information, pure/mixed state Measurement
Lecture 20: entanglement and density operator
Lecture 21: teleportation and super-dense coding
Lecture 22: coherent classical communication
Lecture 23: entropy/mutual information, lossless compression, channel capacity
Lecture 24: von Neumann entropy, Schumacher compression
Lecture 25: entanglement concentration, quantum channel capacities

TARGET AUDIENCE: MATH MAJORS + SOME PHYSICS?
MAYBE REDUCE LECTURES, COMPLEXITY OF TOPICS?

Mathematics for Physics -1 :

Part 1: Linear Algebra: Operators on Finite-Dimensional Vector Spaces (1st May 2026)- Jim, Hasib?, Onirban,

Lecture 01: Vector Spaces - I
Lecture 02: Vector Spaces - II
Lecture 03: Linear Operators - I
Lecture 04: Linear Operators - II
Lecture 05: Eigenvalues and Eigenvectors - I
Lecture 06: Eigenvalues and Eigenvectors - II
Lecture 07: Inner Product Spaces
Lecture 08: Self-adjoint Operators and Spectral Theorem
Lecture 09: Traces and Determinants
Lecture 10: RSA Encryption

Part 2: Operators on Hilbert Spaces

Lecture 01: Finite-dimensional Hilbert Spaces
Lecture 02: Infinite-dimensional Hilbert Spaces
Lecture 03: Bounded Operators
Lecture 04: Unbounded Operators

Lecture 05: Self-adjoint Operators
Lecture 06: Spectrum
Lecture 07: Spectral Theorem - I
Lecture 08: Spectral Theorem -II
Lecture 09: Spectral Theorem - III
Lecture 10: Algebraic and Topological Duals
Lecture 11: Gelfand Triplet and Dirac Bra-ket Formalism
Lecture 12: Position and Momentum Operators in QM
Lecture 13: Canonical Commutation Relations in QM
Lecture 14: One-parameter Group of Unitary Operators
Lecture 15: Spectral theory of a Quantum Harmonic Oscillator
Lecture 16: Trace-class Operators and Partition Function of a Quantum Harmonic Oscillator

Schedule:

Date and time (Bangladesh Standard Time)	Lecture	Lecturer/Speaker
01/05/2026, 7pm-9pm	L01	
02/05/2026, 7pm-9pm	L02	
03/05/2026, 7pm-9pm	L03	
09/05/2026, 7pm-9pm	HW	
15/05/2026, 7pm-9pm	L04	
16/05/2026, 7pm-9pm	L05	
23/05/2026, 7pm-9pm	HW	
30/05/2026, 7pm-9pm	L06	
31/05/2026, 7pm-9pm	L07	
05/06/2026, 7pm-9pm	L08	
06/06/2026, 7pm-9pm	L09	
07/06/2026, 7pm-9pm	L10	
12/06/2026, 7pm-8pm	L11	
13/09/2024, 4pm-5pm	L12	
14/06	HW	
19/06	L13	

20/06	L14	
26/06/2024, 7pm-8pm	L15	
27/06/2024, 7pm-8pm	L16	
28/06	HW	

Mathematics for Physics -2 : Basics of Differential Geometry

Part - I: Basics

Lecture 01: Essentials of Topology - I
Lecture 02: Essentials of Topology - II
Lecture 03: Smooth Manifolds - I
Lecture 04: Smooth Manifolds - II
Lecture 05: Smooth Manifolds with Boundary
Lecture 06: Tangent Bundle
Lecture 07: Vector Fields and Flows
Lecture 08: Differential Forms
Lecture 09: Orientation of a Manifold
Lecture 10: Integration on a Manifold
Lecture 11: Stokes' Theorem

Part - 2: Semi-Riemannian Geometry
Lecture 12: Semi-Riemannian Manifolds
Lecture 13: Connections
Lecture 14: Curvatures
Lecture 15: Geodesics
Lecture 16: Jacobi Vector Fields
Lecture 17: Semi-Riemannian Submanifolds
Lecture 18: Hypersurfaces: (Anti-)de Sitter
Lecture 19: Trigonometry in the Spaces of Constant Curvature
Lecture 20: Time-orientation and Spacestime
Lecture 21: Killing Vector Fields

Lecture 22: Isometry group: Poincaré Group
Lecture 23: Spacetime Singularity
Lecture 24: Example of Black Hole Spacetimes: Schwarzschild
Lecture 25: Example of Cosmological Spacetimes: Robertson-Walker

Mini Course on Basic Mechanics and Special Relativity (March 2026 - April 2026)

Lecture 01: Review of Newtonian Mechanics
Lecture 02: Variational Principle and an Introduction to Lagrangians
Lecture 03: Physical Scenarios and Lagrange Multipliers
Lecture 04: Hamiltonian Dynamics and Physical Scenarios
Lecture 05: Classical Scattering Theory
Lecture 06: Velocity of Propagation of Interaction and Intervals
Lecture 07: Proper Time, Lorentz Transformation and Transformation of Velocities
Lecture 08: Four-Vectors and Four-Dimensional Velocities, Time-Dilation, Length Contraction
Lecture 09: Review of Principle of Least Action and Energy-Momentum (conservation)

Course - 3 (Math)

1. Course 1: May-June 2026
2. Course 2: September -October 2026
3. Course 3: Dec 2026 -Jan 2027

PWF Internships (July-September): Increase Mentors, rules updated (at least a meeting per week and mentees fill out the

form after each meeting like String Mentorship Program). Some slots for students from South Asia*?

Forms for TAs in QM…

Graphics: Walid
PWF Internship Program- ARK
MATH- Onirban + Tasnuva, Tahmid
Phy- Mhai + Shamma, Fahim

PWF BD Team: 1. Nabil Iqbal
2. Mishaal Hai
3. Sayeda Tashnuba Jahan
4. Ahmed Rakin Kamal
5. Onirban Islam
6. Noshin Ferdous Shamma

Student Volunteers:

1. Kazi Fahim Reza Arko
2. Tahmid Shahriar
3. Walid Hasan
4. Fahim haque

ICTP PWF Bangladesh Online Summer Internships (mentorship for ICTP and internship for Bangladesh)

This initiative is a summer online internship (2-3 months and 1 meeting per week/per two weeks) opportunity for Bangladeshi students primarily. This will be conducted by PhD students, postdocs and Professors who are conducting active research in the respective fields and will provide hands-on experience to bachelor and masters students. This internship may lead to bachelor and master thesis and in certain cases to research articles. This initiative will heavily boost the research scene in the country.

Prospective directions:

High Energy Physics Theory:

1. Generalised Symmetries
2. Machine Learning in Physics
3. String Cosmology
4. Various topics in String theory: Black Holes, Higher derivative corrections, compactification..
5. AdS/CFT correspondence
6. Black Hole physics

Condensed Matter Theory:

1. Quantum Information
2. Materials for renewable energy
3. Spintronics

Computer Science:

1. Machine Learning in Physics
2. High Performance Computation
3.

Mathematical Physics:

1. Non-commutative Geometry…

Astrophysics:
1. Prof. Asif U Dowla
2.

People to contact:
PhD students in Europe (ICTP, Durham, Bologna, Masaryk), Mir Faruk, Dr. Onirban Islam, Wasif Ahmed, Arpon Paul, Dr. Ratul Mahanta, Shaikot Jahan Shuvo, Tushar Mitro, Shahnoor, Dr. Rafid Mahbub, Dr. Shadman Salam, Prof. KMB Asad, Dr. Lamima Ahad,

Kiran Adhikary?

List of people and emails:

1. Nabil Iqbal
2. Ahmed Rakin Kamal (ahmedrakinkamaltunok@gmail.com)
3. Shadman Salam (shadman.salam.91@gmail.com)
4. Ratul Mahanta (rmphys27@gmail.com)
5. Tousik Samui (tousiksamui@gmail.com ,
6. Tushar Mitra (tusharmitra007700@gmail.com)
7. Shahnoor Rahman (shahnoor.du.phy@gmail.com)
8. Rafid Mahbub (rmahbub503@gmail.com)
9. Arpon Paul (arponpaul1995@gmail.com)
10. Syeda Lammim Ahad (slahad@uwaterloo.ca)
11. Wasif Ahmed Oishik (wasif.ahmed.bd@gmail.com)
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25. Tanvir Ahmed Masum (tmasum@umass.edu)

26. Shovon Biswas (shovon@phas.ubc.ca)
27. Moinul Hossain Rahat (rahat.moin90@gmail.com)
28. Shafaqat Arifin
29. Arpit Das
30. Subhroneel Chakrabarty
31. Dibya Chakrabarti
32. Naveen Umasankar
33. Hassaan Saleem
34. Rafsanjany Jim
35. Shaikh Saad
36. Sumit Biswas
37. Reza Sabbir Suffian
38. Dwaipayan Debnath
39. Talal Ahmed Chowdhury*

Future Courses - 2 (Quantum Field Theory - 1)

Description of the Program

Criteria: Open to all graduate students and undergraduate students with background in SR and QM.

Dates: September-October, 2025

Phase 1: Foundations of Quantum Field Theory

Lecture 1 - 8 by Dr. Shadman Salam*

Lecture 1 & 2: Introduction to Quantum Field Theory, Klein-Gordon (KG) field, quantization of KG field, Green Functions, Noether's theorem

Lecture 3 & 4: Dirac equation, Dirac field and the Quantization of the Dirac field.

Lecture 5 & 6: Maxwell field (electromagnetic field), local gauge invariance and quantization of the electromagnetic field.

Lecture 7 & 8: Interaction pictures (QED) and Feynman diagrams.

Phase 2: Advanced Quantum Field Theory

Lecture 9-13 by Prof. Nabil Iqbal (Durham University)

Lecture 9: Path integral formulation of Quantum Field Theory.
Lecture 10: Basics of gauge theory (Abelian/Non-Abelian).
Lecture 11:
Lecture 12:
Lecture 13:

Phase 3: Seminars

Seminar 1: Prof. David Tong*

Seminar 2: Dr. Tarek Anous*
Seminar 3: Prof. John Mcgreevy*

Motivation for the Application (Required by the ICTP PWF form):

Bangladesh – a country of 200 million with universities in almost every one of the 64 districts in the country – suffers from the lack of adequate training and research opportunities in physics, particularly along ICTP's HECAP line of research. With the exception of 1-2 universities in the capital city of Dhaka, Bangladeshi universities have very little research or courses in theoretical physics. Major cities like Chittagong, Sylhet, Khulna, Rajshahi, Rangpur, Barisal suffer heavily from the lack of adequate courses and skilled professionals in teaching courses in theoretical physics and training them to become experts and top researchers. The important point to note is that almost all public universities in the country have physics degrees at both the undergraduate and masters level, but the coverage of advanced topics in theoretical physics is low compared to international standards.

We nevertheless believe – based on our experience working with enthusiastic and energetic students – that Bangladesh has the potential to eventually be an international leader in theoretical physics. We will seek to make some steps towards helping it realize its full potential in several ways:

1. We plan to hold two online **introductory lecture courses** a year, at approximately the level of a Masters program. These lectures will cover the two basic courses required to embark in theoretical high-energy physics: one course will cover General Relativity, and the other will cover Quantum Field Theory. It is hoped that this will allow any Bangladeshi senior undergraduate to obtain a basic grounding in these foundational subjects. In 2024 we already held a very successful online QFT school – attended by about 120 students – and we will build on the experience obtained to make this a recurring school.
2. We will hold a yearly **Jamal Nazrul Islam Winter School**, which is an in-person school which will have courses on more advanced topics (e.g. string theory, cosmology, quantum information, etc.) of current interest, usually delivered by visiting high-profile lecturers from abroad.
3. We will organize a **remote internship program**, which will be designed to pair interested undergraduate and Masters students in Bangladesh with more experienced mentors who will guide them through either (a) a reading course in a topic of modern research, or (b) more optimistically, a finite-length research opportunity. The target mentors will be mostly PhD students, postdocs, or faculty members abroad.

A detailed plan follows for each component.

Course Description for Intro to General Relativity

The course on "Introduction to General Relativity" will provide a basic introduction to the field at the level of a usual Masters course in General Relativity. Importantly, there will be a pre-school which gives an overview of important concepts in special relativity.

There will be two halves to the main course. In the first, we will then have approximately 8 2-hour lectures covering tensor calculus, mathematics of curved spacetimes, and the basics of Einstein's theory of relativity, at the level of (e.g.) Sean Carroll's textbook *Spacetime and Geometry: Introduction to General Relativity.* In the second half we will move to more advanced topics; we will provide an introduction to the theory of black holes and gravitational waves, and the basic FRW theory of cosmology, ending with a brief summary of inflation. Following the model established by our online QFT school, there will be tutorials and homework assignments, as well as guest seminars on topics of research interest from leaders in the field.

Possible lecturers for school: Mr. Hasaan Saleem, Mir Mehedi Faruk

Possible guest lecturers: Daniel Baumann, Cliff Burgess, Alejandra Castro, Damian Galante, Atish Dabholkar

Course Description for Intro to QFT:

To be modified slightly:

This project is aimed at final year undergraduate and masters students from Bangladesh as we have basically no standard quantum field theory course being run in the whole country and the situation of high energy physics research in the country is sub-standard. We plan to run an online school in three phases where we start from students having exposure to quantum mechanics and special relativity and take them through the journey where they will develop the tools of quantum field theory and learn about some directions beyond the standard model. This school will prepare them for graduate studies and will develop the base for doing research in high energy physics even if they reside in the country.

Possible lecturers for school: Nabil Iqbal, Tin Sulejmanpasic, Daniele Dorigoni
Possible guest lecturers: David Tong, Tarek Anous

Online Internship Program

This initiative is an online internship (2-3 months and 1 meeting per week/per two weeks) opportunity for Bangladeshi students primarily. This will consist either of a guided reading program or (more optimistically) a program of research, where the latter will involve a computational component. We envisage that this will be conducted by PhD students, postdocs and professors who are conducting active research in the respective fields and will provide hands-on experience to bachelor and masters students. We believe that it will not be hard to find researchers – particularly at the more junior level – who are willing to supervise one or two students per summer at this level, and we will seek to match students to interested supervisors.

Some possible topics for guided reading programs at this level include:

Generalized global symmetries, inflation models inspired by string theory, string cosmology, black holes in extra dimensions, elementary topics in the gauge/gravity correspondence, etc.

Some possible topics for a research-based internship involve:

Machine learning techniques for physics, Monte Carlo simulations for generalized global symmetries and phases, non-reciprocal phase transitions, etc.

This internship will lead to increased connections between students and the international research community; in cases where the program goes well (and with the cooperation of Bangladeshi institutes) it could lead to bachelor and master thesis, and even more optimistically to research articles.

Things to do:

1. Email potential people for the internship- Nabil
2. Post the year planner - urgent, ARK, Tasnuva, Mishaal
3. Post GRC poster and form- Emailing Natasa: Rakin
4. Gather emails: Tasnuva, Mishaal
5. Email Cliff Burgess, Daniel Baumann and Samir Mathur? : Nabil
6. Make a schedule for GR

Bangladesh - a country of 200 million with universities in almost every one of the 64 districts in the country - suffers from the lack of adequate training and research opportunities in physics. With the exception of 1-2 universities in the capital city of Dhaka, Bangladeshi universities have very little research or courses in theoretical physics. Major cities like Chittagong, Sylhet, Khulna, Rajshahi, Rangpur, Barisal suffer heavily from the lack of adequate courses and skilled professionals in teaching courses in theoretical physics and training them to become experts and top researchers. The important point to note is that almost all public universities in the country have physics degrees at both the undergraduate and masters level, but the coverage of advanced topics in theoretical physics is low compared to international standards.
We nevertheless believe - based on our experience working with enthusiastic and energetic students - that Bangladesh has the potential to eventually be an international leader in theoretical physics. We will seek to make some steps towards helping it realize its full potential in several ways:
1. We plan to hold some elementary and advanced schools from early 2026 to March-2027 . These lectures will cover the some basic courses required to embark in a career in physics: Specifically one course will cover quantum mechanics and quantum information, another on linear algebra followed by differential geometry and finally a course on lagrangian mechanics and special relativity. It is hoped that this will allow any Bangladeshi undergraduate to obtain a basic grounding in these foundational subjects. In 2024 and 2025 we already held a very successful online QFT school and GR school.
2. Following last year's success, we will organize a remote internship program, which will be designed to pair interested undergraduate and masters students in Bangladesh with more experienced mentors who will guide them through either (a) a reading course in a topic of modern research, or (b) more optimistically, a finite-length research opportunity. The target mentors will be mostly PhD students, postdocs, or faculty members abroad.